Gradient-Based Adversarial Examples with Focus on the Fast Gradient Sign Method (FGSM) and its Extensions

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CISPA
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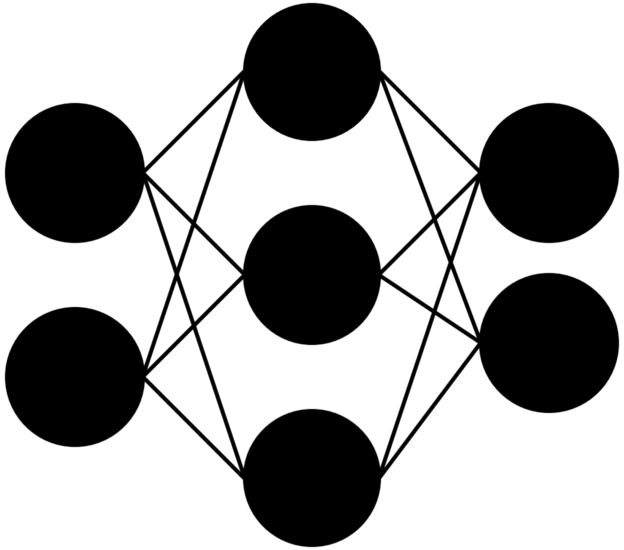
Lecture Plan on Adversarial Examples

1. **Introduction to Adversarial Examples:** imperceptibly perturbed inputs that fool machine learning (ML) models.
2. **FGSM (Fast Gradient Sign Method):** generate an adversarial example with a single gradient step.
3. **Extensions of FGSM:** BIM (Basic Iterative Method) and PGD (Projected Gradient Descent).
4. **Robust Models**: adversarial training against adversarial examples.

Predictions of Machine Learning Models



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Adversarial Examples: a Bug in ML

[Szegedy et al. ‘13], [Biggio et al. ‘13], [Goodfellow et al. ’14], …

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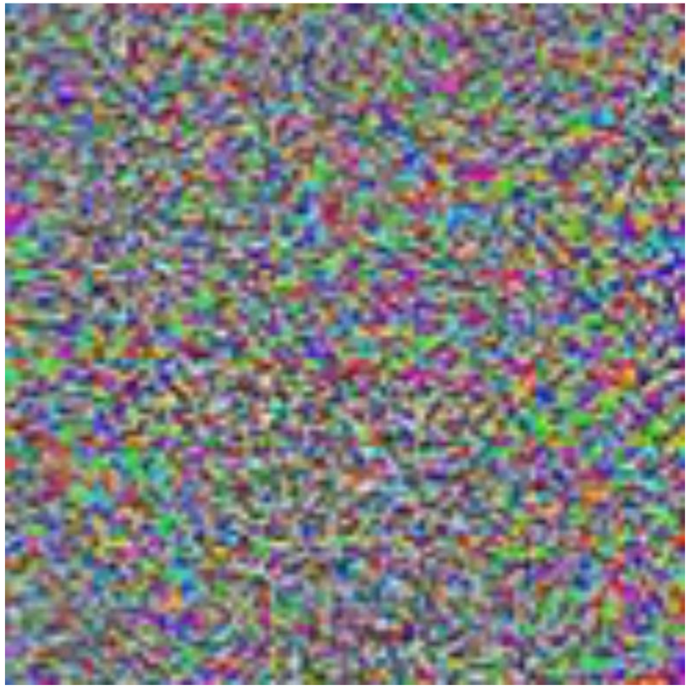

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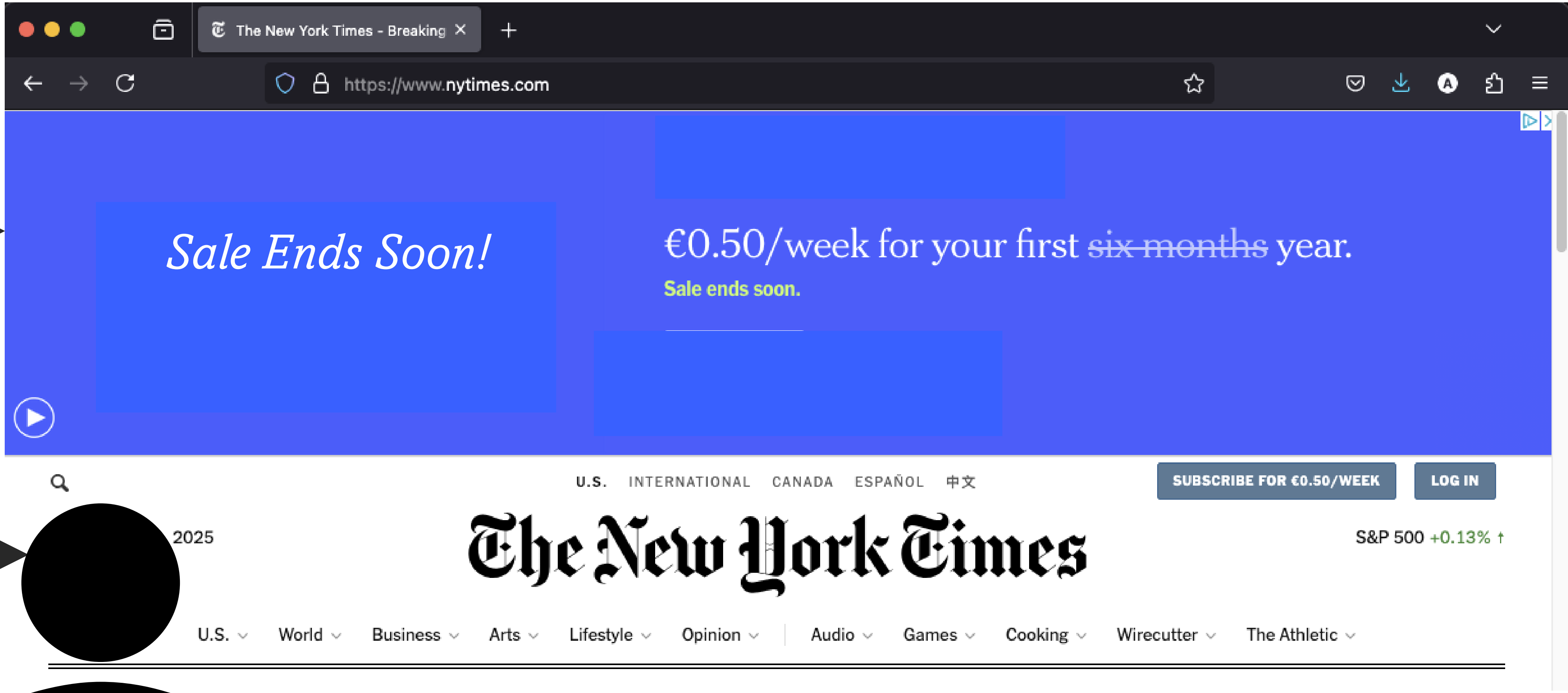
88% Tabby Cat | Adversarial Noise | 100% Guacamole

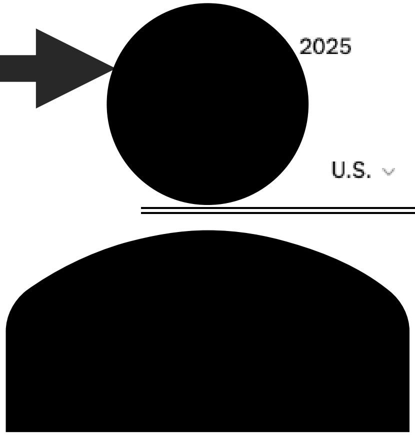
Adversarial Examples Fool Ad-Blockers

[Tramèr et al. '19]

Publishers show ads to users



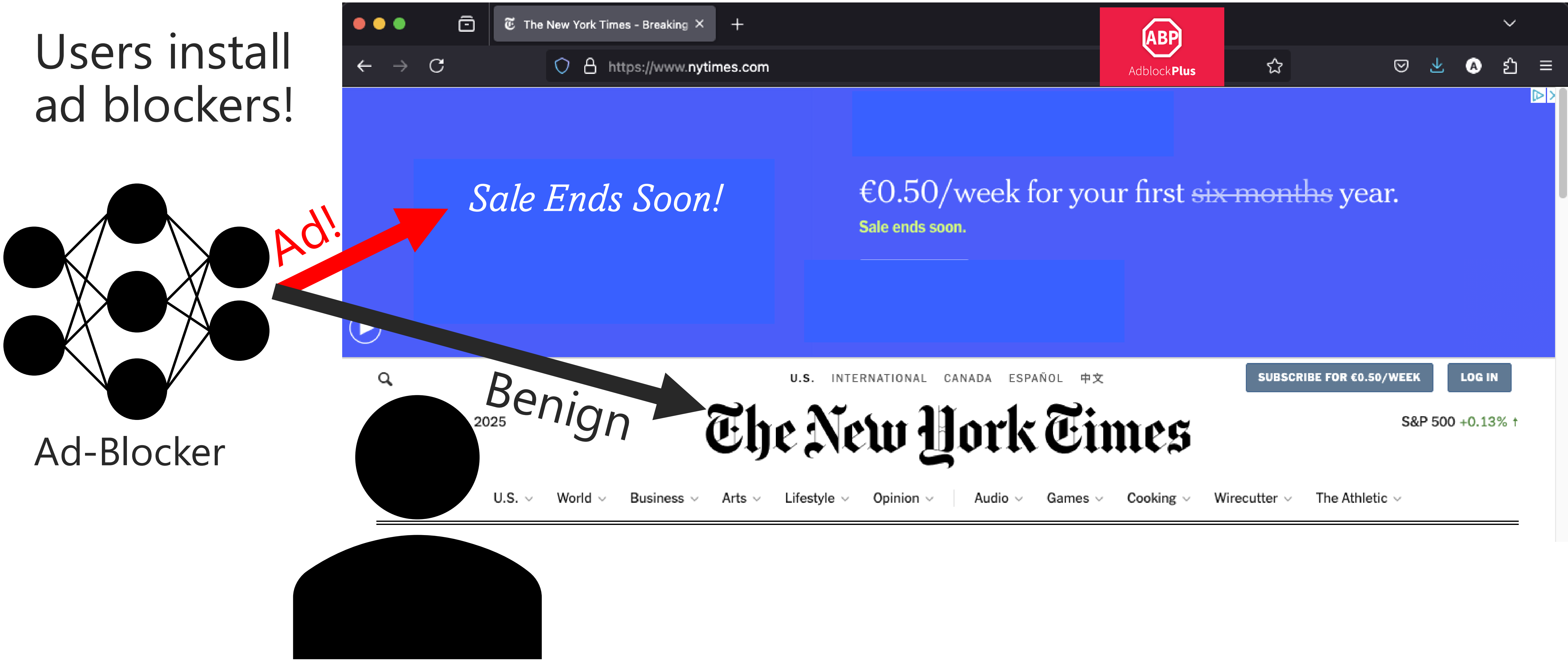
Users don't want to see ads!



Adversarial Examples Fool Ad-Blockers

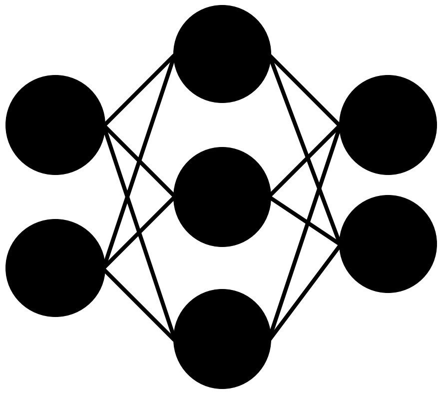
[Tramèr et al. '19]

Users install ad blockers!

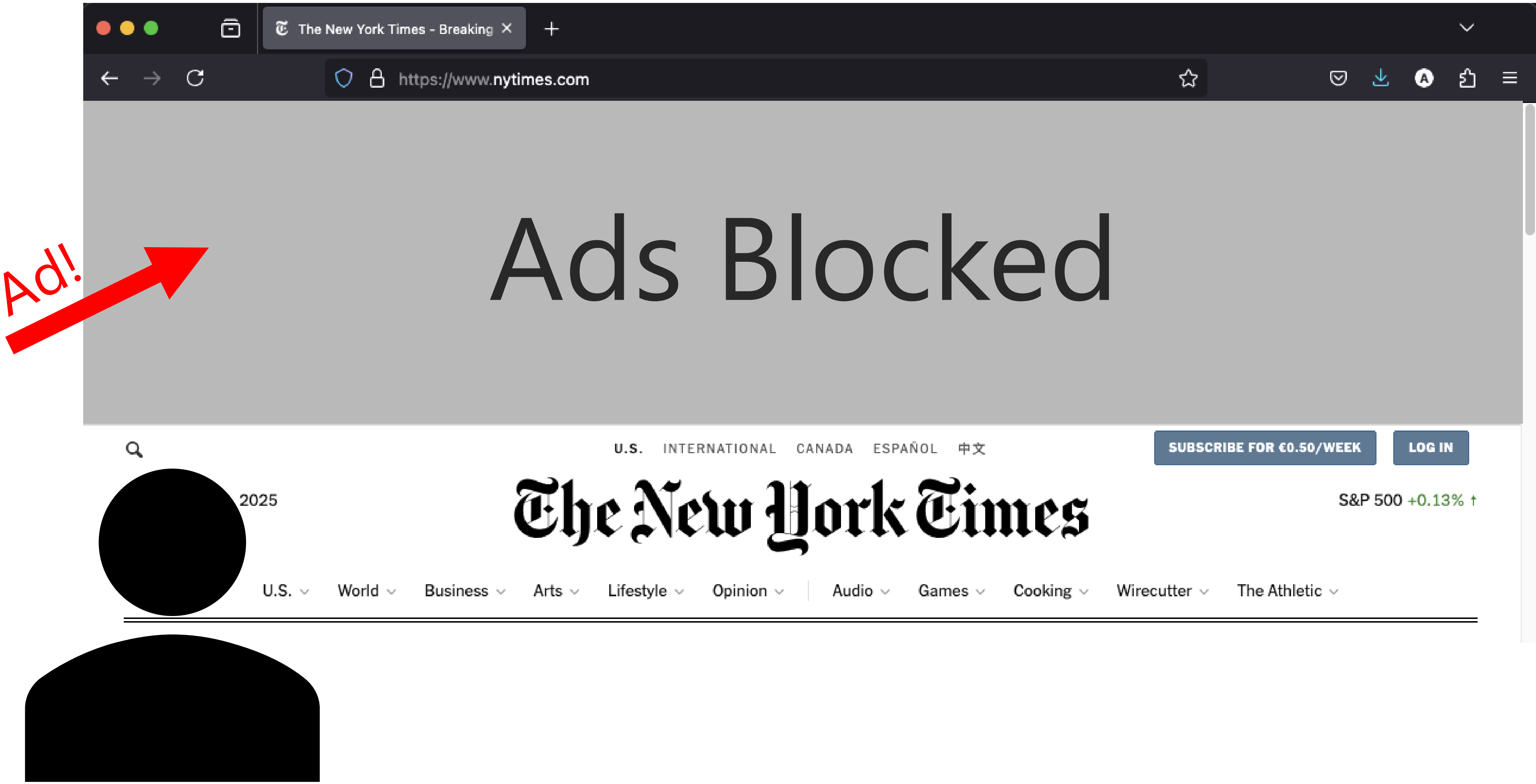


Adversarial Examples Fool Ad-Blockers

[Tramèr et al. '19]



Ad-Blocker



Adversarial Examples Fool Ad-Blockers

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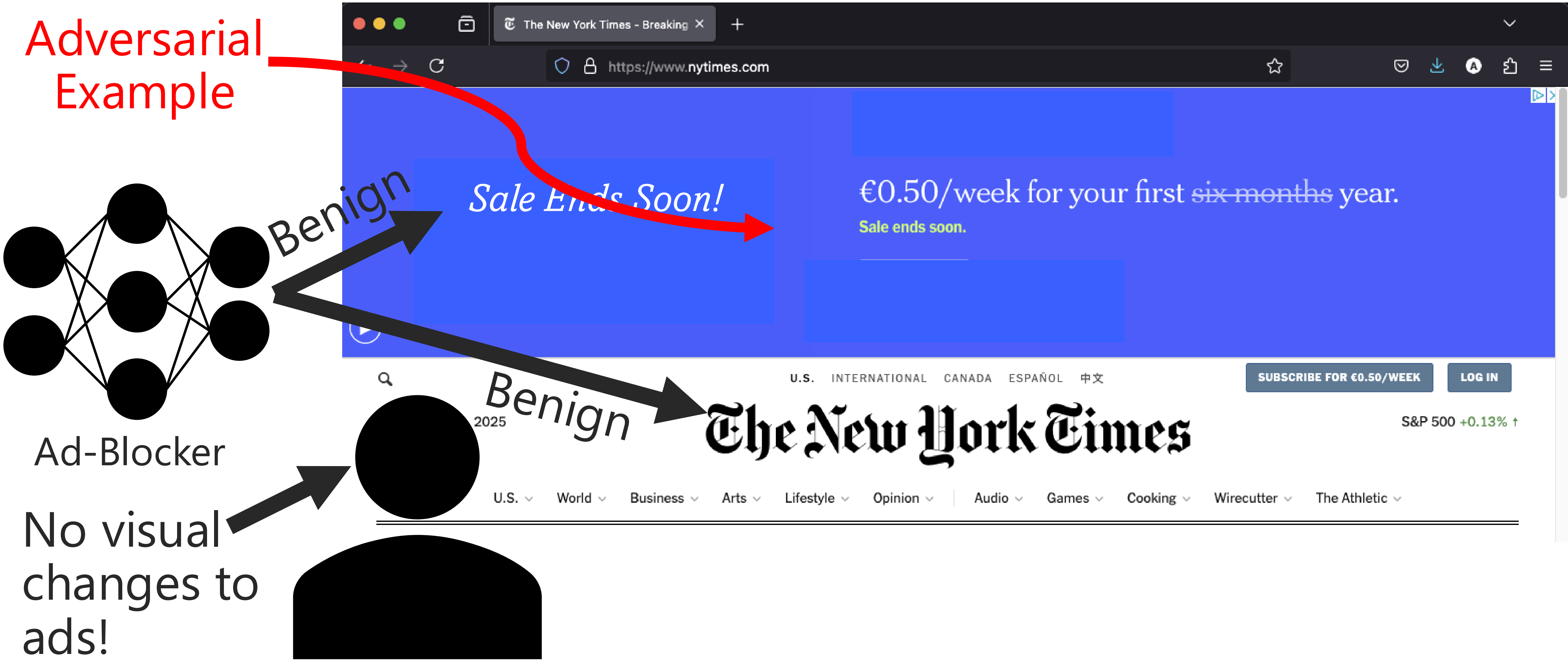

Imperceptible Adversarial Examples

[Goodfellow et al. '14]

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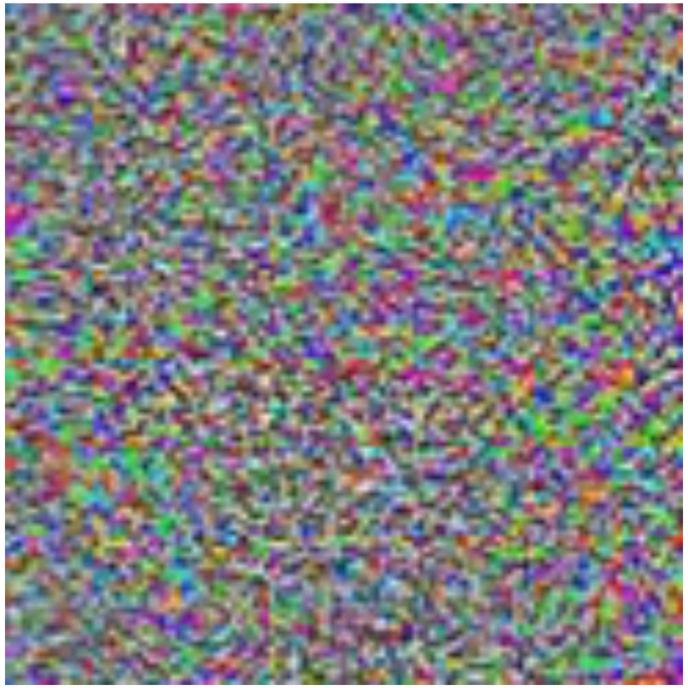


Adversarial Noise

$\leq \boldsymbol{\varepsilon}$

Imperceptible Adversarial Examples

[Goodfellow et al. '14]

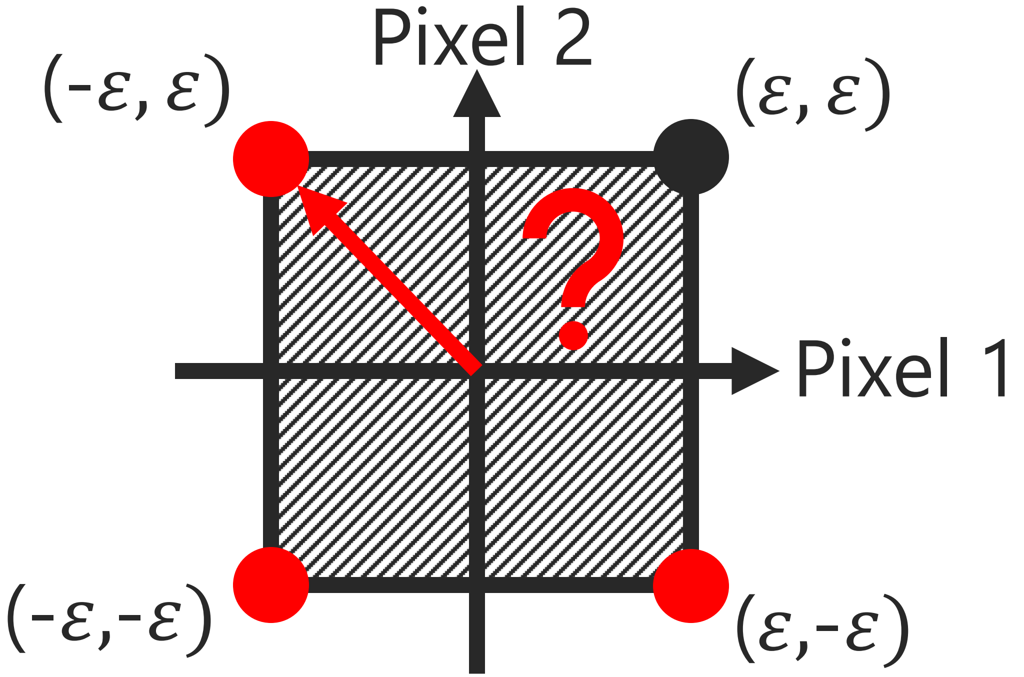
Adversarial Noise

$\leq \boldsymbol{\varepsilon}$

Max-Norm: $\|v\|_\infty := \max_i |v_i|$

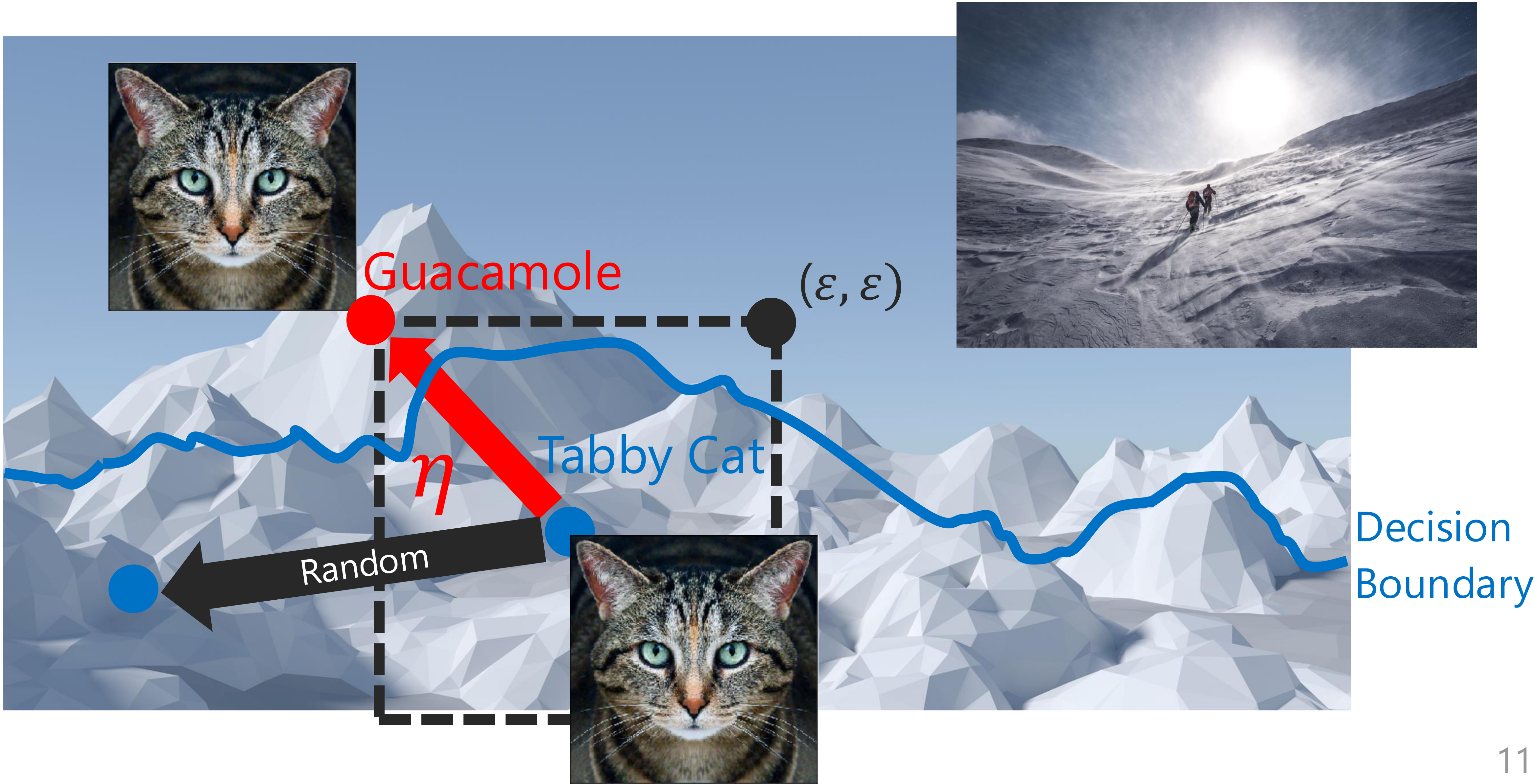
$\left\| \begin{array}{|c|c|} \hline -2 & 1 \\ \hline \end{array} \right\|_\infty = 2$

$\|{\color{red}\eta}\|_\infty \leq \varepsilon$

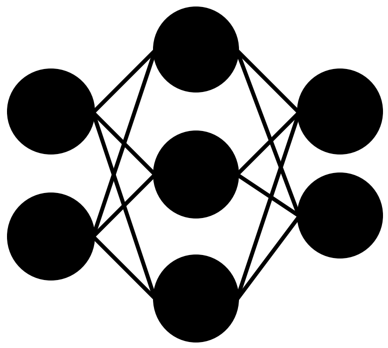


Adversarial Examples via Gradient Ascent

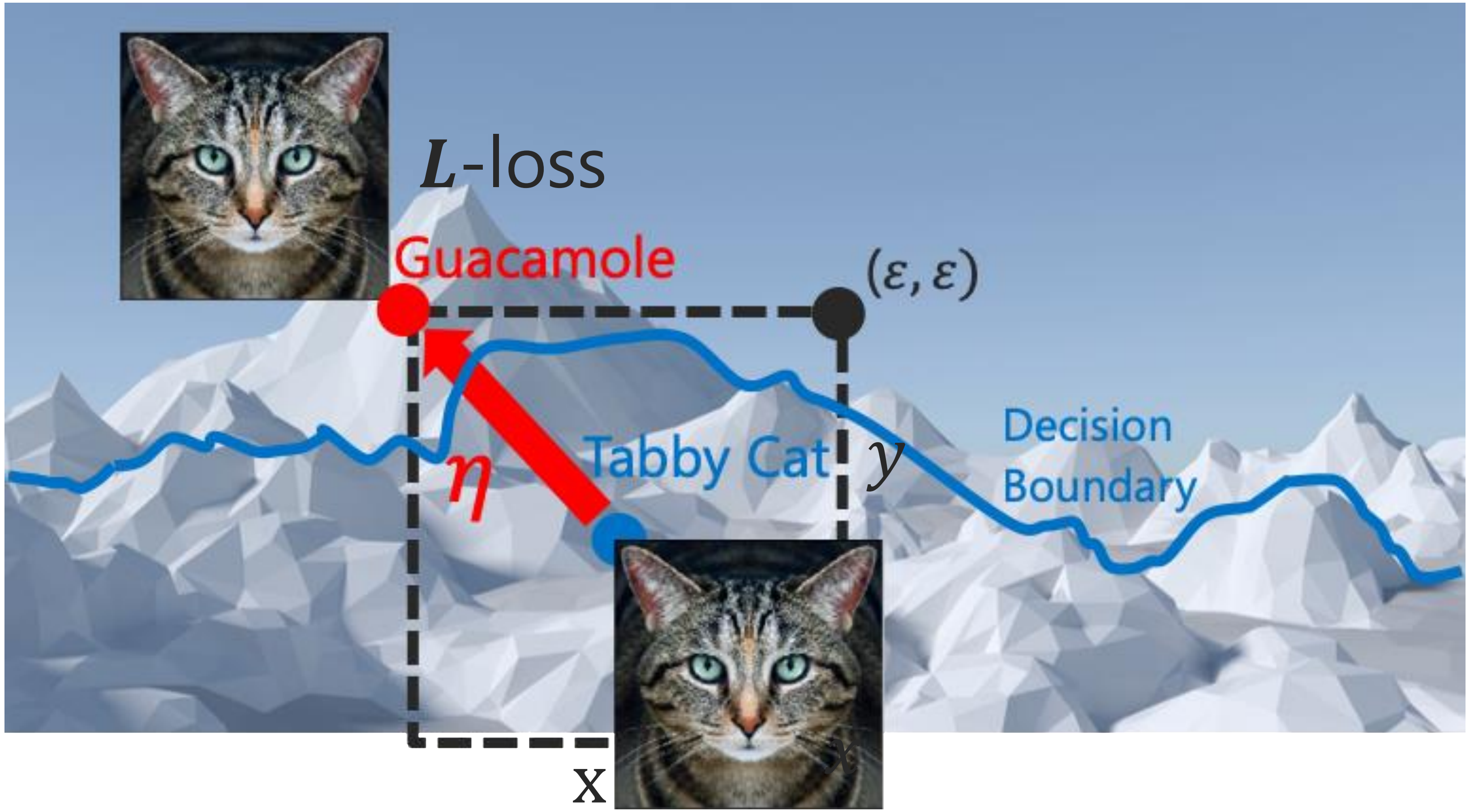
[Tramèr, ’21]



Adversarial Examples via Gradient Descent



$\boldsymbol{\theta}$-model parameters



Gradient for the adversarial noise: $\eta = \nabla_{\mathrm{x}} \mathrm{L}(\mathrm{x}, \mathrm{y}, \theta)$

via backpropagation through the model $\boldsymbol{\theta}$ to input x

How to Maximize & Scale the Noise?

[Goodfellow et al. '14]

$\boldsymbol{\nabla_x L(x, y, \theta)}$

show direction

$$\mathrm{sign}(x) = \begin{cases} -1 \text{ if } x < 0 \\ 0 \text{ if } x = 0 \\ 1 \text{ if } x > 0 \end{cases}$$

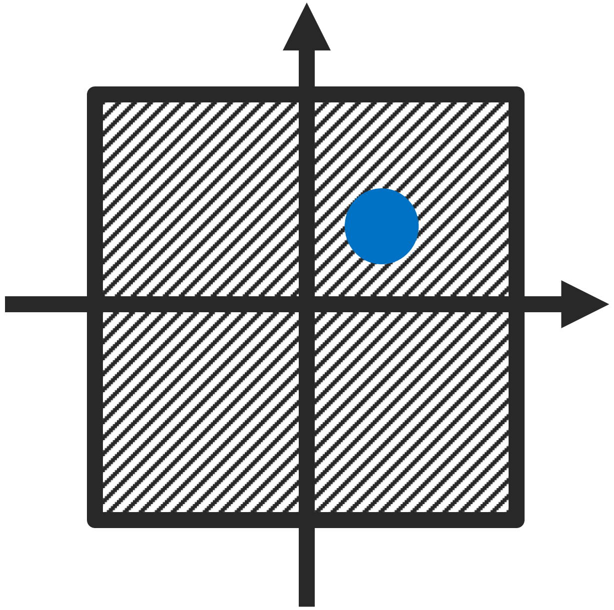
Scale

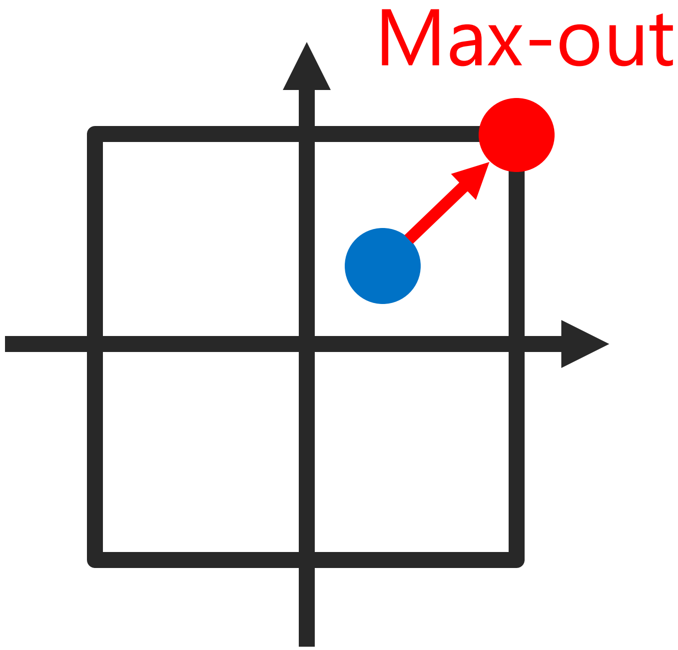
ε sign$(\nabla_x L(x, y, \theta))$

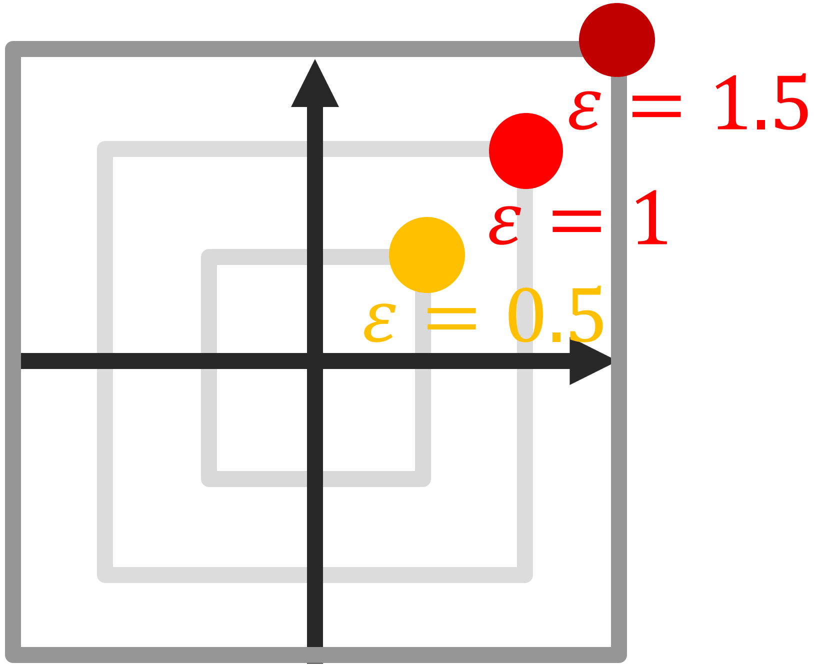
Bound

$\|\nabla_x L(x, y, \theta)\|_\infty \leq \varepsilon$

sign$(\nabla_x L(x, y, \theta))$







Fast Gradient Sign Method (FGSM)

[Goodfellow et al. '14]

The threat model:

1. Pixel-wise restriction: $d(x, x^*) = \|x - x^*\|_\infty = \max_i |x_i - x_i^*| \leq \varepsilon$
2. White-box: full access to the target network, including gradients.

Finds adversarial examples via **linearizing** the training loss:

$$\max_{x^*:\|x^*-x\|_\infty \leq \varepsilon} L(f(x^*), y) \approx L(f(x), y) + (x^* - x) \cdot \nabla_x L(f(x), y))$$

FGSM uses **sign(·)** to max out and meet the max-norm constraint:

$$x^* = x + \varepsilon \operatorname{sign}(\nabla_x L(x, y, \theta))$$

Fast Gradient Sign Method (FGSM)

[Goodfellow et al. '14]

Generate **adversarial example** x^* with a single gradient step:

$$x^* = x + \varepsilon\,\mathrm{sign}(\nabla_x L(x, y, \theta))$$

x – input, ε – perturbation magnitude, ∇_x - gradient, y – label, L – loss, θ – model parameters

$$\mathrm{sign}(x) = \begin{cases} -1 \text{ if } x < 0 \\ \;\;0 \text{ if } x = 0 \\ \;\;1 \text{ if } x > 0 \end{cases}$$

Untargeted vs Targeted Adversarial Examples

[Kurakin et al. '16]

Untargeted: increase the loss value for **the true label** $\boldsymbol{y_{true}}$

$$x^* = x + \varepsilon \operatorname{sign}(\nabla_x L(x, \boldsymbol{y_{true}}, \theta))$$

Targeted: maximize probability of **a target class** $\boldsymbol{y_{target}}$

$$x^* = x - \varepsilon \operatorname{sign}(\nabla_x L(x, \boldsymbol{y_{target}}, \theta))$$

Extensions of Fast Gradient Sign Method

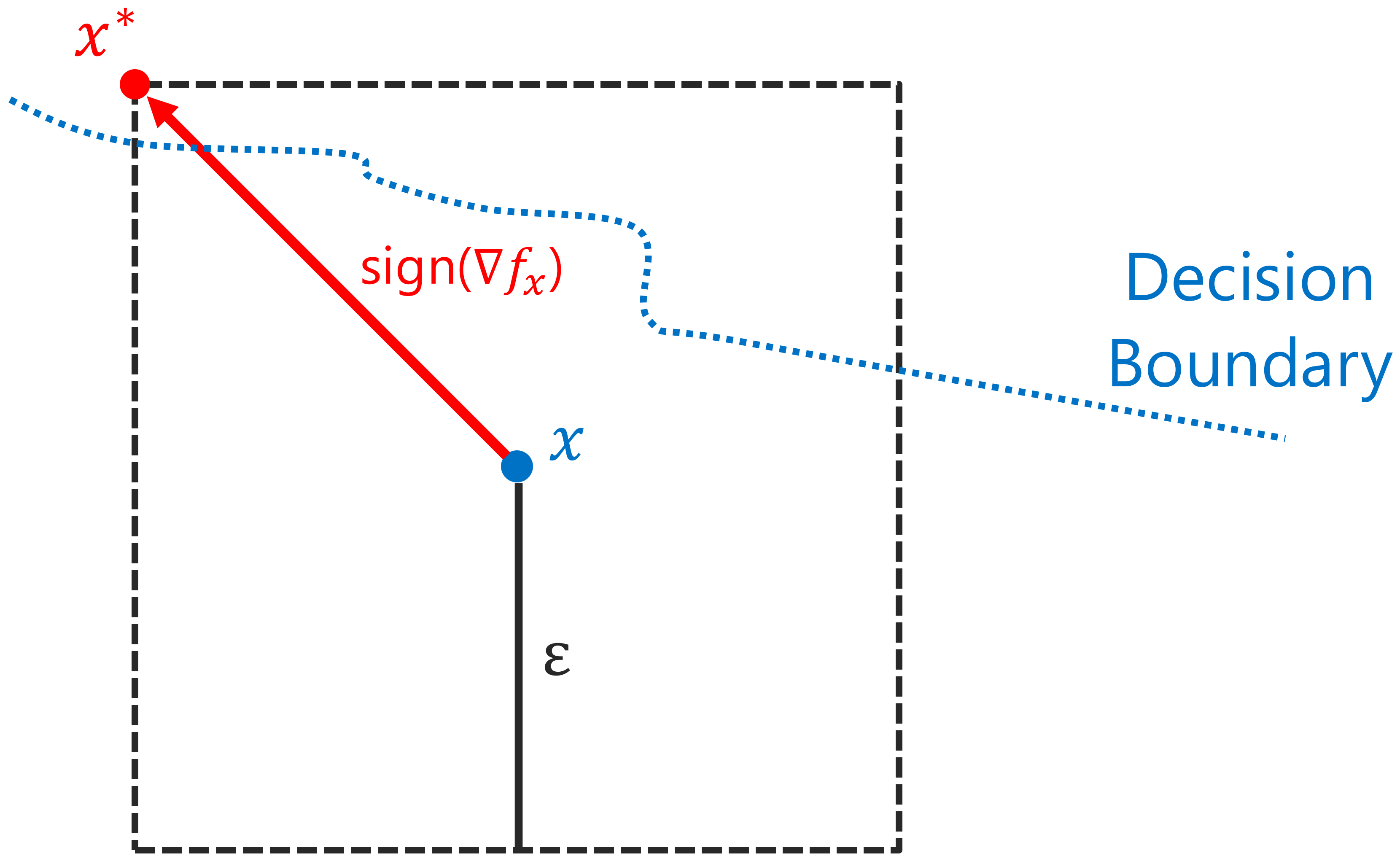
Fast Gradient Sign Method (FGSM)-generate adversarial examples with a single gradient step. [Goodfellow et al. 14]

$$x^* = x + \varepsilon\, \mathrm{sign}(\nabla_x L(x, y, \theta))$$

Why do we need the extensions of FGSM?

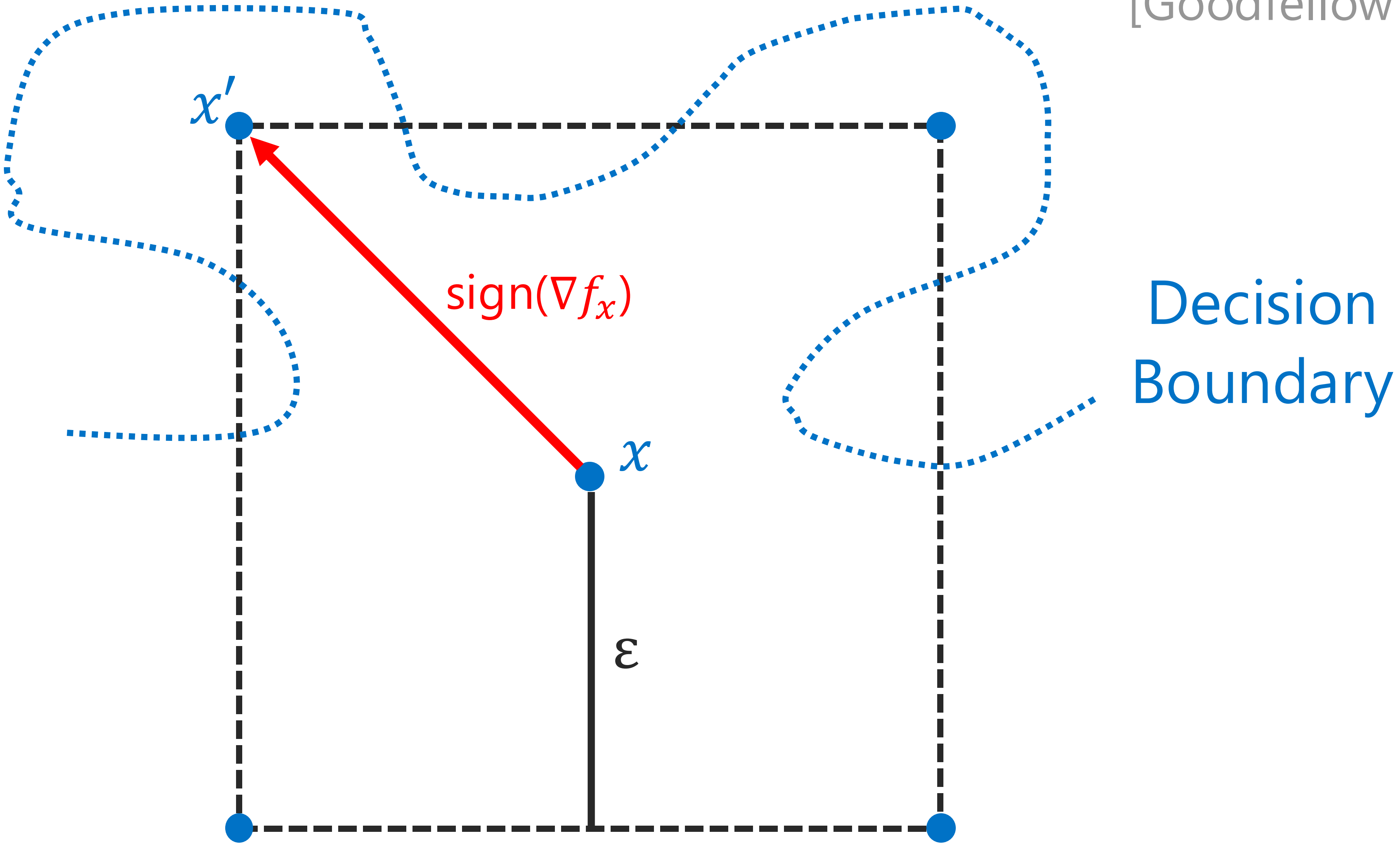
How Does FGSM Find Adversarial Examples?

[Goodfellow et al. '14]



FGSM Fails on Complex Decision Boundaries

[Goodfellow et al. '14]



Extensions of Fast Gradient Sign Method

Fast Gradient Sign Method (FGSM)-generate adversarial examples with a single gradient step. [Goodfellow et al. '14]

$$x^* = x + \varepsilon\,\mathrm{sign}(\nabla_x L(x, y, \theta))$$

[Kurakin et al. '16]

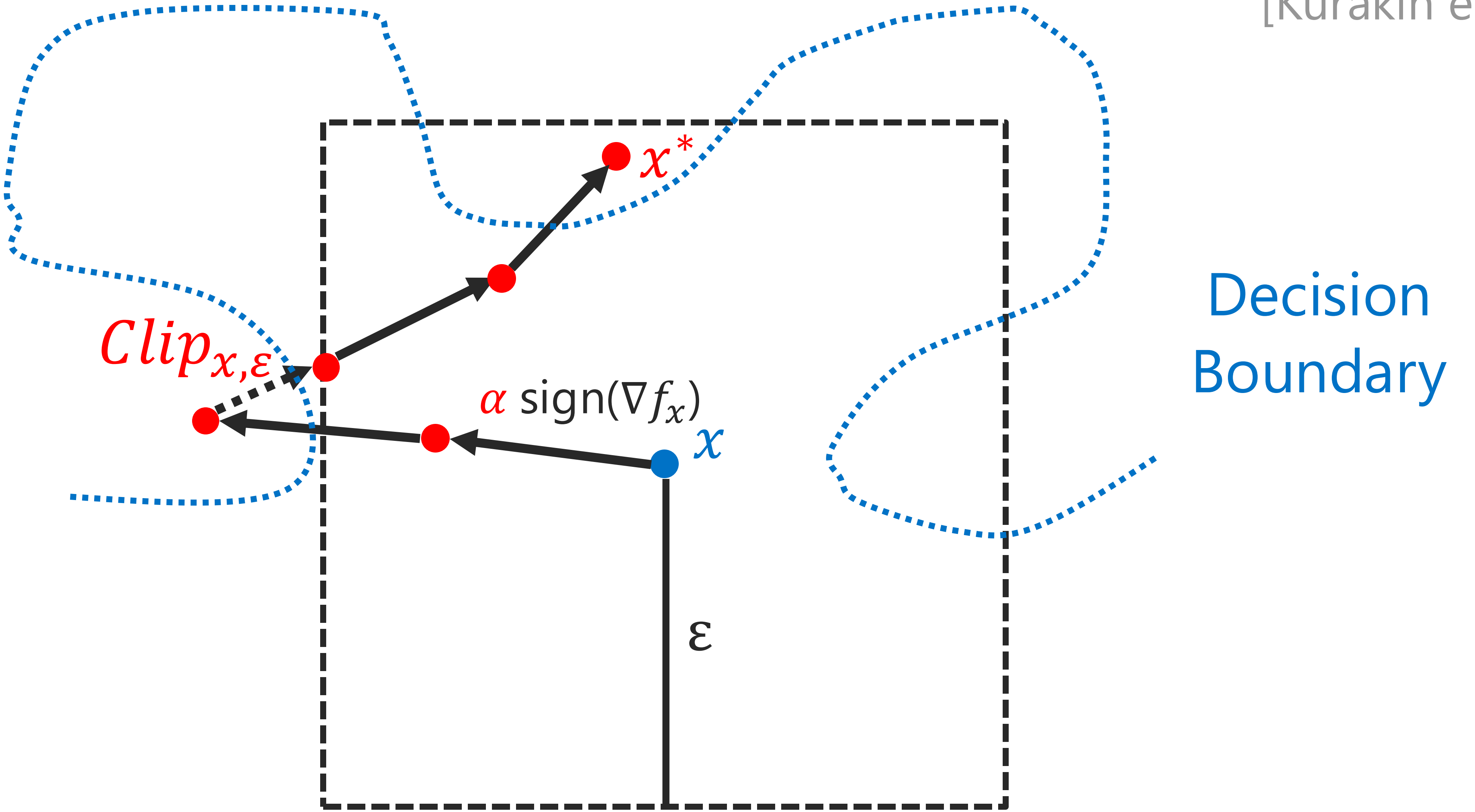
Basic Iterative Method (BIM) - multiple small FGSM steps:

for k in $[0, n]$:

$$x^* = Clip_{x,\varepsilon}\left(x^* + \alpha\,\mathrm{sign}\left(\nabla_{x^*} L(x^*, y, \theta)\right)\right)$$

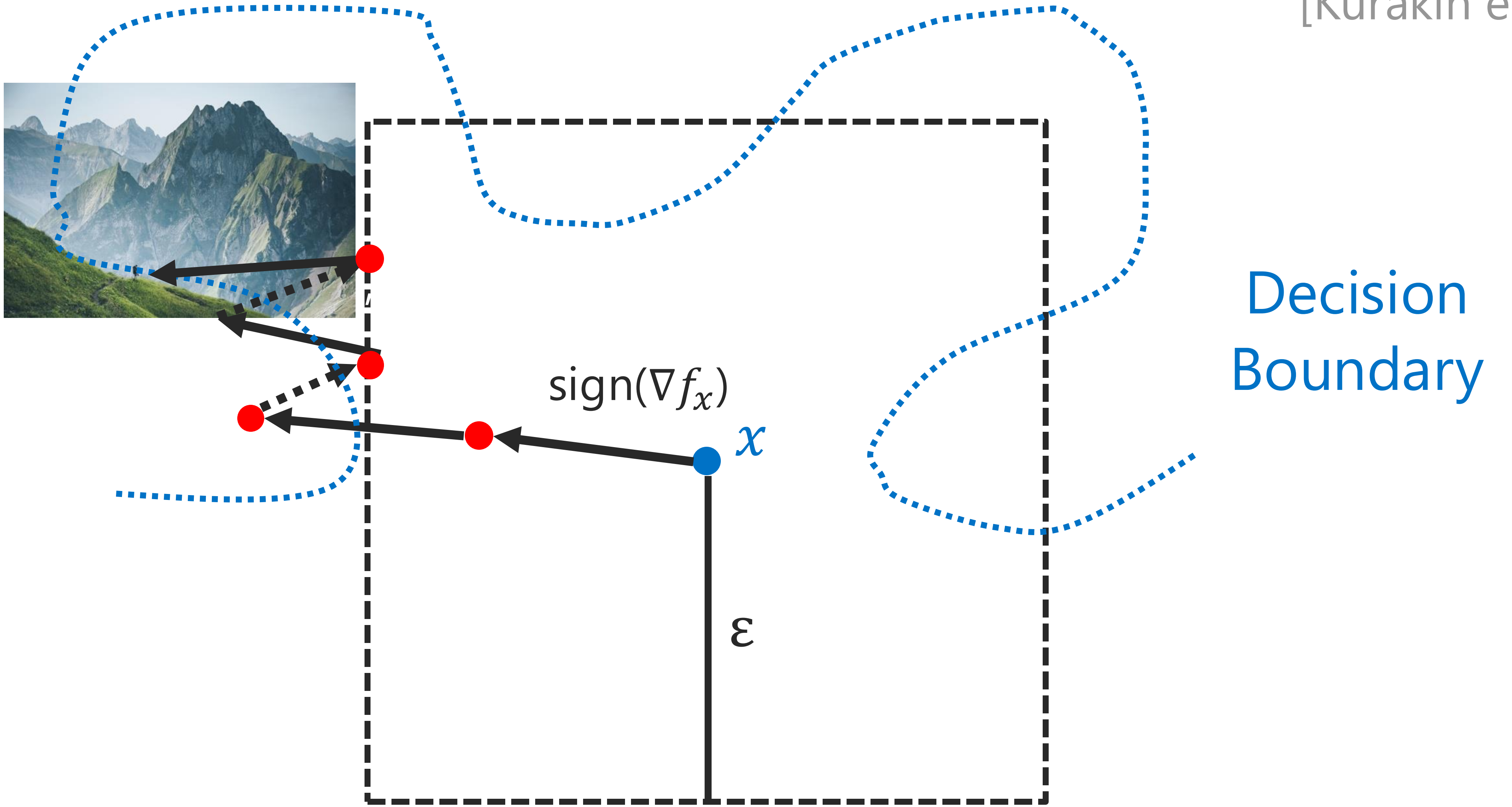
How Does BIM Find Adversarial Examples?

[Kurakin et al. '16]



BIM Stuck in Local Optima

[Kurakin et al. '16]



Extensions of Fast Gradient Sign Method

Fast Gradient Sign Method (FGSM)-generate adversarial examples with a single gradient step. [Goodfellow et al. '14]

$$x^* = x + \varepsilon\,\mathrm{sign}(\nabla_x L(x, y, \theta))$$

[Kurakin et al. '16]

Basic Iterative Method (BIM) - multiple k small FGSM steps:

for k in $[0, n]$:

$$x^* = Clip_{x,\varepsilon}\left(x^* + \alpha\,\mathrm{sign}\left(\nabla_{x^*} L(x^*, y, \theta)\right)\right)$$

[Mądry et al. '17]

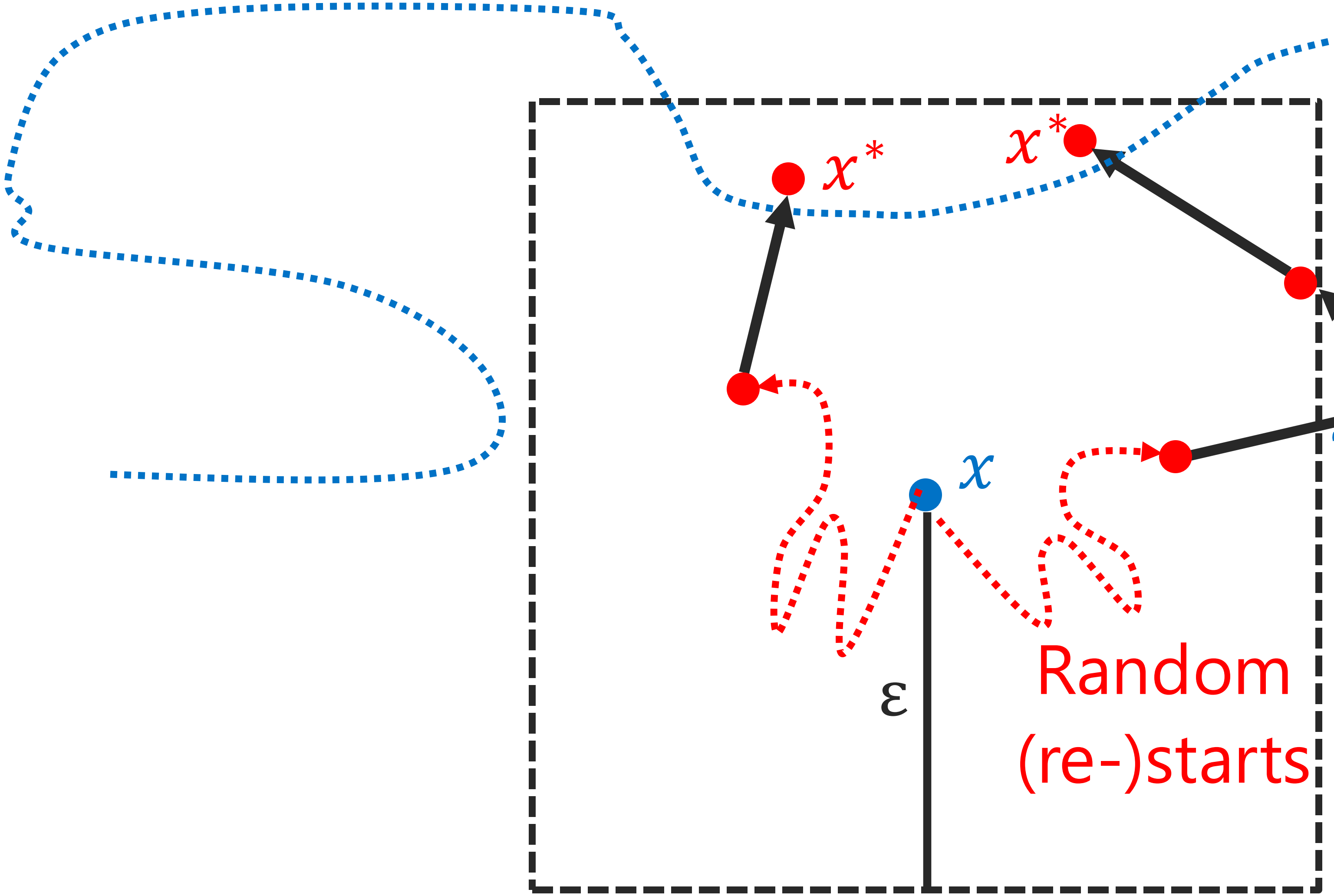
Projected Gradient Descent (PGD) – random (re-)starts

$$x^* = x + Uniform(-\varepsilon, \varepsilon)$$

and the basic iterative method.

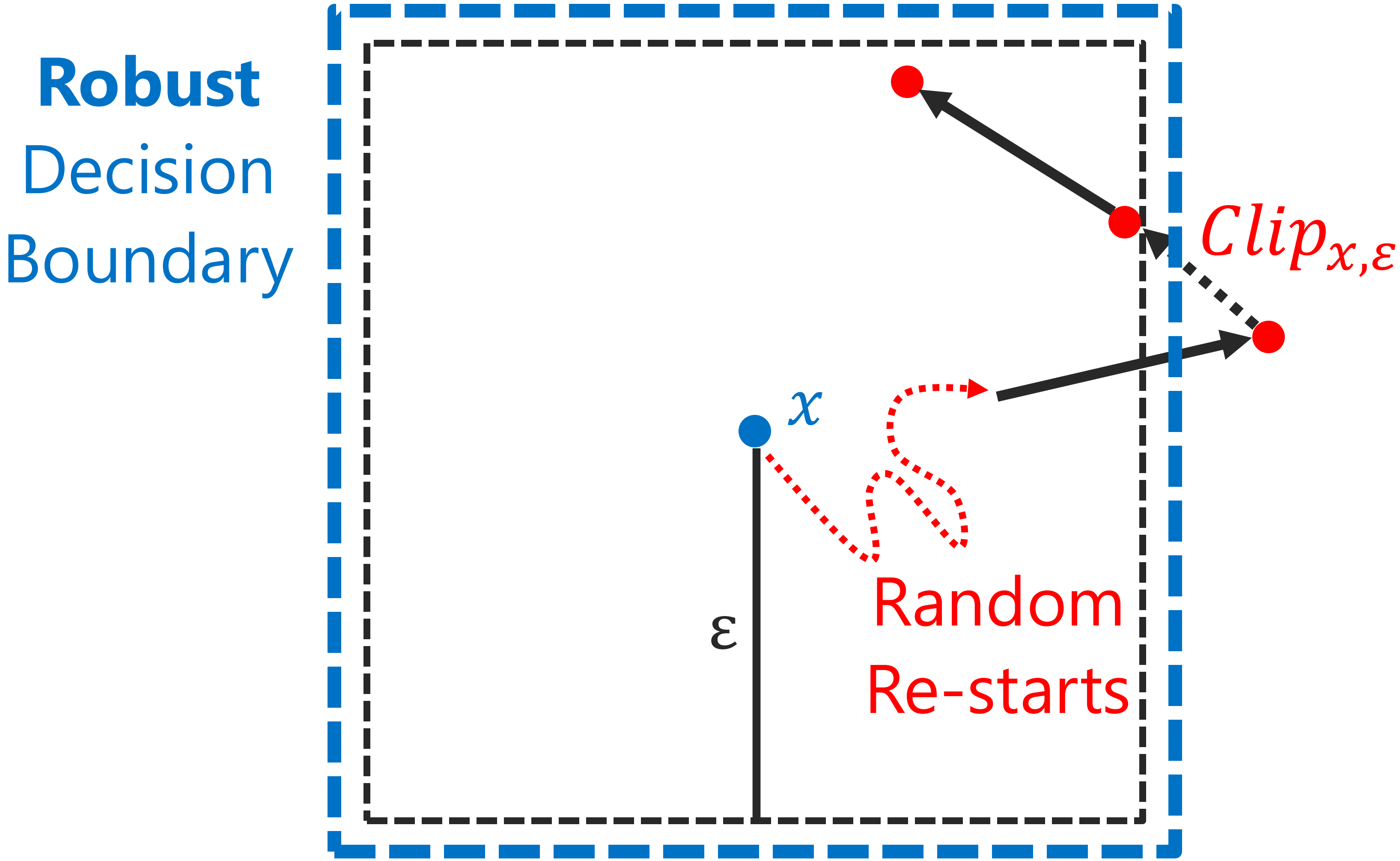
Hoe Does PGD Find Adversarial Examples?

[Mądry et al. ‘17]



PGD Designed for Adversarial Training

[Mądry et al. ‘17]



Robust Models via Adversarial Training

[Mądry et al. '17]

1. Traditional training: minimize the loss with respect to θ

$$\min_{\theta} E_{(x,y)\sim D}[L(x, y, \theta)]$$

Robust Models via Adversarial Training

[Mądry et al. '17]

1. Traditional training: minimize the loss with respect to θ

$$\min_{\theta} E_{(x,y)\sim D}[L(x, y, \theta)]$$

2. Adversarial example: maximize the loss using x and η

$$\max_{\|\eta\|_\infty \leq \varepsilon} L(x + \eta, y, \theta)$$

Robust Models via Adversarial Training

[Mądry et al. '17]

1. Traditional training: minimize the loss with respect to θ

$$\min_{\theta} E_{(x,y)\sim D}[L(x, y, \theta)]$$

2. Adversarial example: maximize the loss using x and η

$$\max_{\|\eta\|_\infty \leq \varepsilon} L(x + \eta, y, \theta)$$

3. Adversarial training: minimize adversarial risk during training

$$\boxed{\min_{\theta} E_{(x,y)\sim D}\left[\max_{\|\eta\|_\infty \leq \varepsilon} L(x + \eta, y, \theta)\right]}$$

Gradient-Based Adversarial Examples

1. **Adversarial Examples:** imperceptible perturbation of inputs that fool ML models.

2. **FGSM (Fast Gradient Sign Method):** creates adversarial examples in a single gradient step.

3. **Extensions of FGSM: BIM** iteratively performs many small FGSM steps and **PGD** further performs random restarts.

4. **Robust Models**: adversarial training incorporates the adversarial examples into the standard training process.

Materials & Prompts for the Discussion

Advanced Lecture:

https://bit.ly/4jEZjbx

Reading List:

https://bit.ly/4cHENoA

1. Are there attacks in ℓ_1 or ℓ_2 norms (apart from max-norm)?
2. What is the cost of adversarial training?
3. Which types of adversarial examples should be used for adversarial training and which of them are the most effective?
4. Are bigger models more robust and why?
5. Can we preserve high clean accuracy for robust models?

Advanced Lecture:

https://bit.ly/4jEZjbx

Reading List:

https://bit.ly/4cHENoA

Thank you!

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Course on Trustworthy Machine Learning

Bibliography

[BS14] Joan Bruna, Christian Szegedy, Ilya Sutskever, Ian Goodfellow, Wojciech Zaremba, Rob Fergus, Dumitru Erhan, "Intriguing properties of neural networks", ICLR 2014.

[GS15] Ian Goodfellow, Jonathon Shlens, Christian Szegedy, "Explaining and Harnessing Adversarial Examples", ICLR 2015.

[HX15] Ruitong Huang, Bing Xu, Dale Schuurmans, Csaba Szepesvari, "Learning with a Strong Adversary", arXiv 2015.

[PM16] Nicolas Papernot, Patrick McDaniel, Ian Goodfellow, "Transferability in Machine Learning: from Phenomena to Black-Box Attacks using Adversarial Samples", arXiv 2016.

[GP17] Ian Goodfellow and Nicolas Papernot, "Is attacking machine learning easier than defending it?", cleverhans blog post 2017.

[KG17] Alexey Kurakin, Ian Goodfellow, Samy Bengio, "Adversarial Machine Learning at Scale", ICLR 2017.

[KGB17] Alexey Kurakin, Ian Goodfellow, Samy Bengio, "Adversarial examples in the physical world", arXiv 2017.

[PM17] Nicolas Papernot, Patrick McDaniel, Ian Goodfellow, Somesh Jha, Z. Berkay Celik, Ananthram Swami, "Practical Black-Box Attacks against Machine Learning", ASIA CCS 2017.

[AC18] Anish Athalye, Nicholas Carlini, David Wagner, "Obfuscated Gradients Give a False Sense of Security: Circumventing Defenses to Adversarial Examples", ICML 2018.

[AE18] Anish Athalye, Logan Engstrom, Andrew Ilyas, Kevin Kwok, "Synthesizing Robust Adversarial Examples", ICML 2018.

[CW18] Nicholas Carlini, David Wagner, "Audio Adversarial Examples: Targeted Attacks on Speech-to-Text", arXiv 2018.

[MM18] Aleksander Mądry, Aleksandar Makelov, Ludwig Schmidt, Dimitris Tsipras, Adrian Vladu, "Towards deep learning models resistant to adversarial attacks", ICLR 2018.

[SS18] Ludwig Schmidt, Shibani Santurkar, Dimitris Tsipras, Kunal Talwar, Aleksander Madry, "Adversarially Robust Generalization Requires More Data", NeurIPS 2018.

[TK18] Florian Tramèr, Alexey Kurakin, Nicolas Papernot, Ian Goodfellow, Dan Boneh, Patrick McDaniel, "Ensemble Adversarial Training: Attacks and Defenses", ICLR 2018.

[CR19] Jeremy Cohen, Elan Rosenfeld, Zico Kolter, "Certified Adversarial Robustness via Randomized Smoothing", ICML 2019.

[CR19] Jeremy Cohen, Elan Rosenfeld, Zico Kolter, "Certified Adversarial Robustness via Randomized Smoothing", ICML 2019.

[GG19] Chuan Guo, Jacob Gardner, Yurong You, Andrew Gordon Wilson, Kilian Weinberger, "Simple Black-box Adversarial Attacks", ICML 2019.

[HL19] Dan Hendrycks, Kimin Lee, Mantas Mazeika, "Using Pre-Training Can Improve Model Robustness and Uncertainty", ICML 2019.

[LA19] Mathias Lecuyer, Vaggelis Atlidakis, Roxana Geambasu, Daniel Hsu, Suman Jana, "Certified Robustness to Adversarial Examples with Differential Privacy", IEEE 2019.

[LC19] Bai Li, Changyou Chen, Wenlin Wang, Lawrence Carin, "Certified Adversarial Robustness with Additive Noise", NeurIPS 2019.

[RK19] Kevin Roth, Yannic Kilcher, Thomas Hofmann "The Odds are Odd: A Statistical Test for Detecting Adversarial Examples", ICML 2019.

[SB19] Lukas Schott, Jonas Rauber, Matthias Bethge, Wieland Brendel, "Towards the first adversarially robust neural network model on MNIST", ICLR 2019.

[SM19] Ali Shafahi, Mahyar Najibi, Mohammad Amin Ghiasi, Zheng Xu, John Dickerson, Christoph Studer, Larry S. Davis, Gavin Taylor, Tom Goldstein, "Adversarial training for free!", NeurIPS 2019.

[TB19] Florian Tramer, Dan Boneh, "Adversarial Training and Robustness for Multiple Perturbations", NeurIPS 2019.

[UB19] Jonathan Uesato, Jean-Baptiste Alayrac, Po-Sen Huang, Robert Stanforth, Alhussein Fawzi, Pushmeet Kohli, "Are Labels Required for Improving Adversarial Robustness?", arXiv 2019.

[YG19] Dong Yin, Raphael Gontijo Lopes, Jon Shlens, Ekin Dogus Cubuk, Justin Gilmer, "A Fourier Perspective on Model Robustness in Computer Vision", NeurIPS 2019.

[CH20] Francesco Croce, Matthias Hein, "Reliable evaluation of adversarial robustness with an ensemble of diverse parameter-free attacks", ICML 2020.

[CR20] Yair Carmon, Aditi Raghunathan, Ludwig Schmidt, John C. Duchi, Percy S. Liang, "Unlabeled Data Improves Adversarial Robustness", NeurIPS 2020.

[GQ20] Sven Gowal, Chongli Qin, Jonathan Uesato, Timothy Mann, Pushmeet Kohli, "Uncovering the Limits of Adversarial Training against Norm-Bounded Adversarial Examples", arXiv 2020.

[IS20] Andrew Ilyas, Shibani Santurkar, Dimitris Tsipras, Logan Engstrom, Brandon Tran, Aleksander Madry, "Adversarial Examples Are Not Bugs, They Are Features", NerurIPS 2020.

[RW20] Leslie Rice, Eric Wong, Zico Kolter, "Overfitting in adversarially robust deep learning", ICML 2020.

[SE20] Lea Schönherr, Thorsten Eisenhofer, Steffen Zeiler, Thorsten Holz, Dorothea Kolossa, "Imperio: Robust Over-the-Air Adversarial Examples for Automatic Speech Recognition Systems", ACSAC 2020.

[TB20] Florian Tramèr, Jens Behrmann, Nicholas Carlini, Nicolas Papernot, Jörn-Henrik Jacobsen, "Fundamental Tradeoffs between Invariance and Sensitivity to Adversarial Perturbations", ICML 2020.

[WR20] Eric Wong, Leslie Rice, J. Zico Kolter, "Fast is better than free: Revisiting adversarial training", ICLR 2020.

[AI21] "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale", under submission to ICLR 2021.

[GA21] "Geometry-aware Instance-reweighted Adversarial Training", under submission to ICLR 2021.

[IA21] "Improving Adversarial Robustness via Channel-wise Activation Suppressing", under submission to ICLR 2021.

Questions?

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Course on Trustworthy Machine Learning

Why does FGSM use the L_∞-norm?

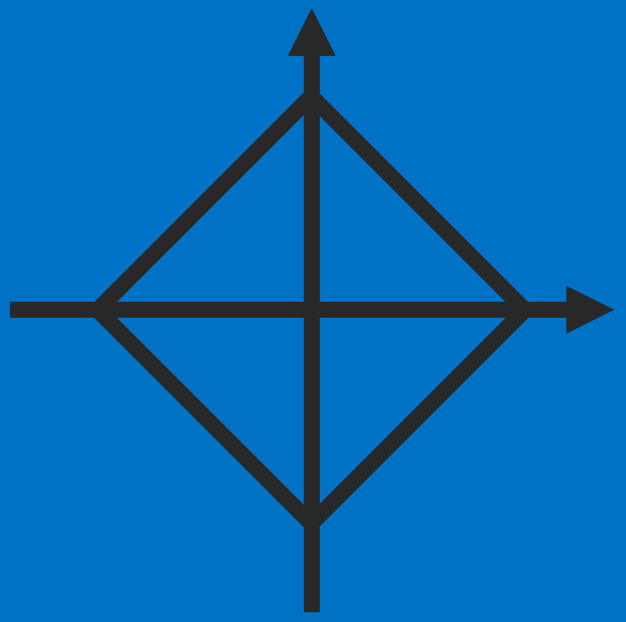
1. **Gives a fast and easy-to-implement attack**: produces a strong attack with a single step by using only the sign of the gradient.
2. **Maximally perturbs each pixel within the ε-ball:** gives an intuitive bound on how much an input is altered.
3. **Aligns well with the structure of images:** controls per-pixel changes and reflects real-world constraints.

Detailed comparison between norms:

1. L_0-norm constraints how many pixels change.
2. L_1-norm constraints the sum of all changes.
3. L_2-norm constraints the magnitude (energy) of all changes.
4. L_1 and L_2 allow some features to change a lot and other stay small.
5. L_∞-norm constraints the maximum allowed change per each pixel.

Norms of Vectors: Unit Ball $\varepsilon = 1$ Examples

$$\|x\|_1 := \sum_{i=1}^{n} |x_i| = 1$$

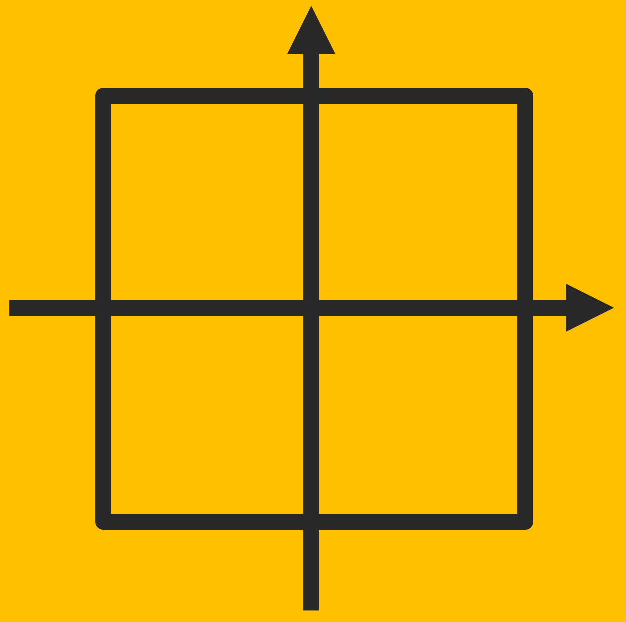


Sum of Changes

$$\|x\|_2 := \left(\sum_{i=1}^{n} |x_i|^2\right) = 1$$

Magnitude of Changes

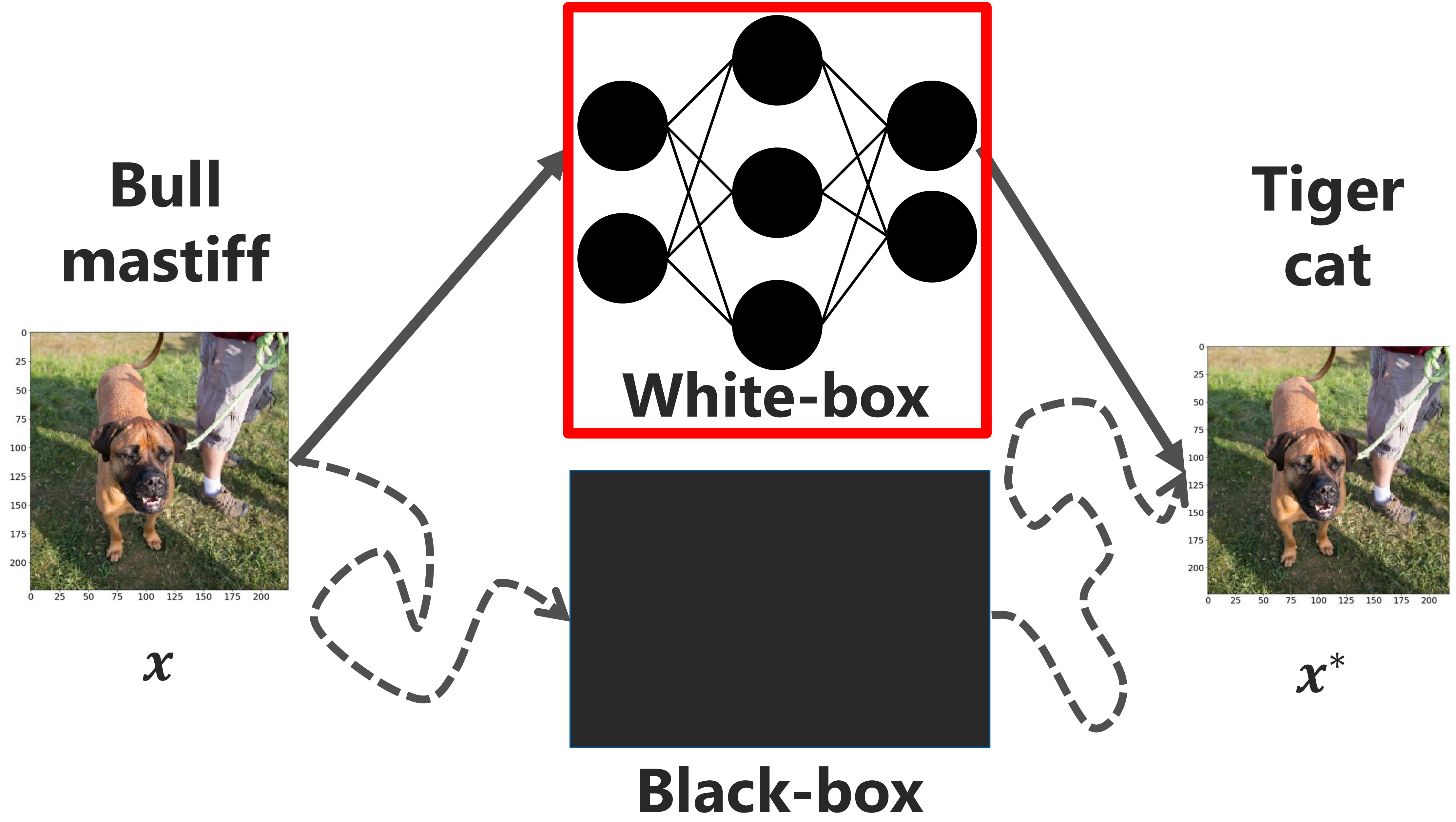
$$\|x\|_\infty := \max_i |x_i| = 1$$



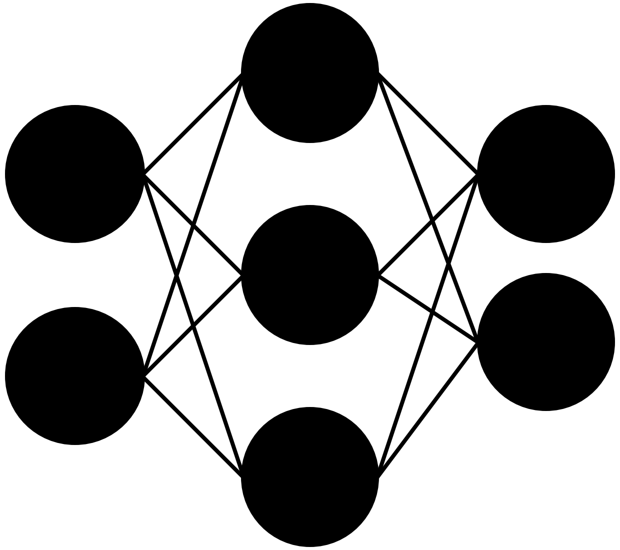
Changes per Pixel

Threat Model

[PM17]



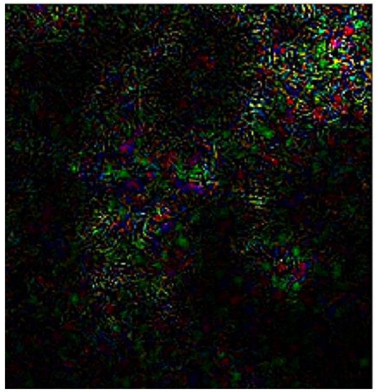
Crafting White-box Attacks



Bull mastiff

Tiger cat

$+$



$=$

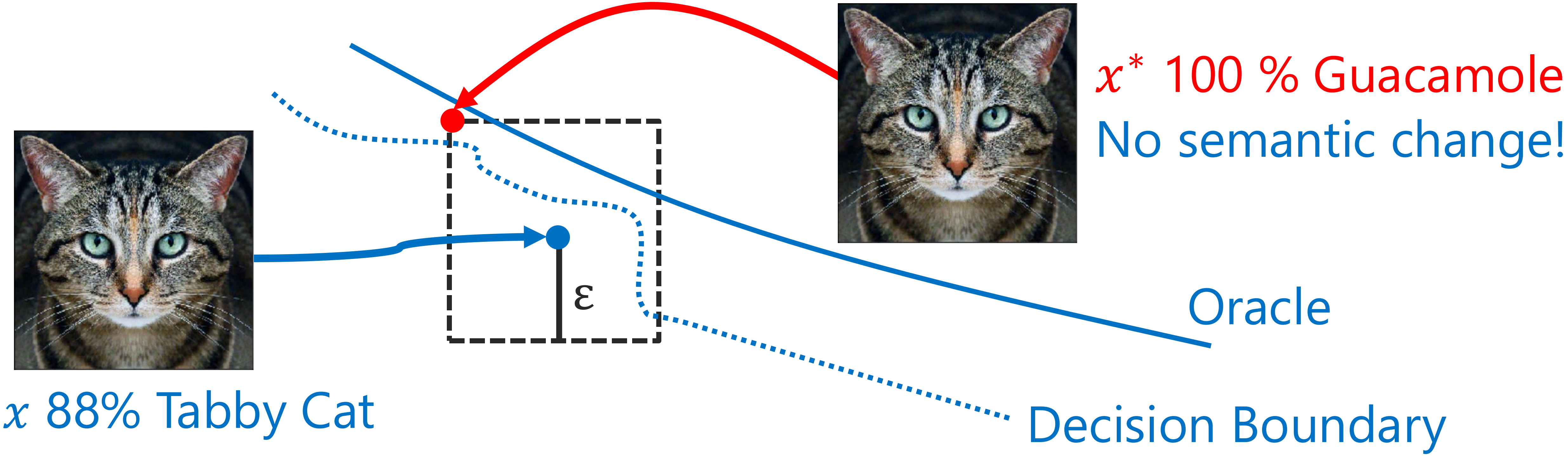
x

Noise: η

x^*

Visualize Adversarial Examples: **ε**-bounded

[Goodfellow et al. '14]



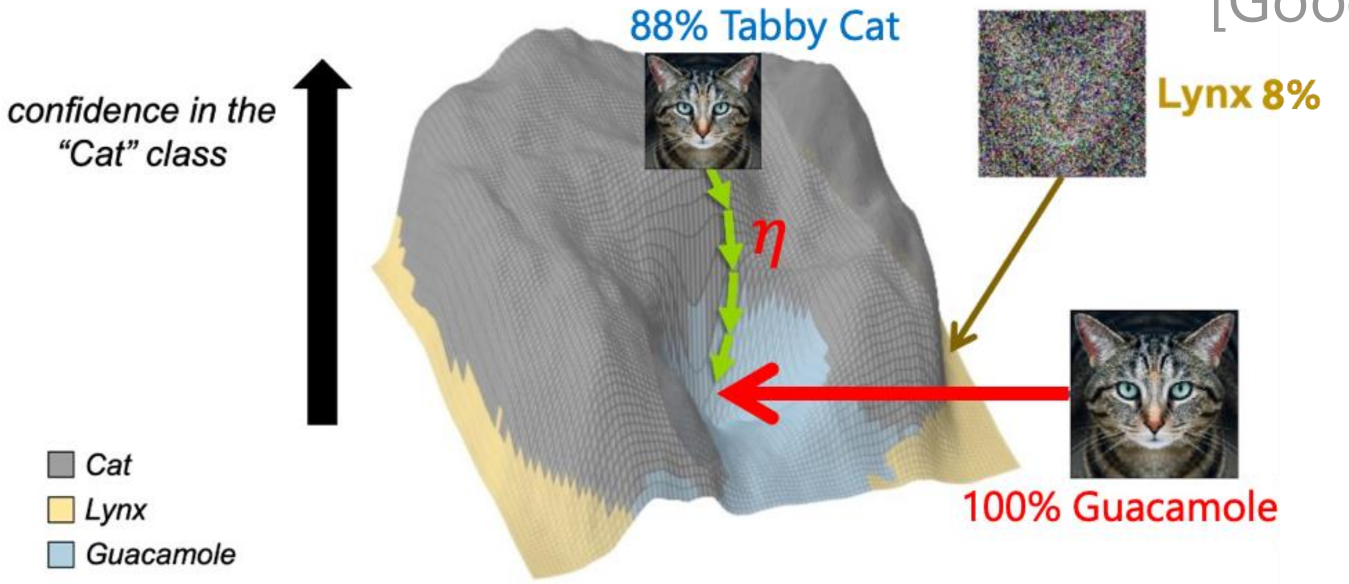
Max Perturbation: $\boldsymbol{\eta} = \|x^* - x\|_\infty \leq \varepsilon$

Model Prediction: $f(x) \neq f(x^*)$

Oracle: $O(x^*) = O(x)$

Linear Explanation of Adversarial Examples

[Goodfellow et al. '14]



Adversarial Example: $\eta = \nabla f_x \rightarrow \|\eta\|_\infty \leq \varepsilon$

Neural Nets ≈ Linear: $f(x) = \theta^T x \rightarrow \nabla_x f = \theta$

Adversarial Activation: $\theta^T \eta$

Maximize the Activation: $\eta = \text{sign}(\nabla_x f)$